WORKED SOLUTION

STEP II 2012, Question 4

Pure • Logarithmic Series Inequalities

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/12-s2-q4

The key idea

Everything comes from the series for $\ln(1+x)$ and the fact that e^t is increasing. To compare $(1+\frac{1}{n})^n$ with e, take logarithms: we need to compare $n\ln(1+\frac{1}{n})$ with 1. An alternating series with shrinking terms can be bracketed in pairs to show it is less than its first term; and the combination $\ln\frac{1+x}{1-x}$ kills all the negative terms, so it is bigger than its first term. Squeezing between the two bounds gives the limit.

Solution

Part (i)

Since $n > 1$ and $k \geq 1$, we have $(k+1)n > n > 1$, and certainly $(k+1)n > k$. Multiplying by $n^k > 0$,

$$(k+1)n^{k+1} > kn^k > 0,$$

and taking reciprocals of these positive numbers reverses the inequality:

$$\frac{1}{(k+1)n^{k+1}} < \frac{1}{kn^k}.$$

Now put $x = \frac{1}{n}$ in the series; this is allowed because $n > 1$ gives $0 < x < 1$. Grouping the terms in pairs after the first,

$$\ln\left(1+\frac{1}{n}\right) = \frac{1}{n} - \left(\frac{1}{2n^2} - \frac{1}{3n^3}\right) - \left(\frac{1}{4n^4} - \frac{1}{5n^5}\right) - \cdots.$$

Each bracket has the form $\frac{1}{kn^k} - \frac{1}{(k+1)n^{k+1}}$ with k even, and so is strictly positive by the inequality just proved.

? **Ask yourself** Are we allowed to bracket an infinite series like this?

Yes, because the series converges. The partial sums with an odd number of terms form a subsequence of all the partial sums, so they tend to the same limit, $\ln(1+\frac{1}{n})$. The bracketed series is exactly the sequence of these odd partial sums.

The partial sums of the bracketed series are $\frac{1}{n} - b_1$, $\frac{1}{n} - b_1 - b_2$, ..., where every $b_j > 0$. They decrease, so all of them are at most $\frac{1}{n} - b_1$, and therefore so is their limit:

$$\ln\left(1+\frac{1}{n}\right) \leq \frac{1}{n} - b_1 < \frac{1}{n}, \qquad b_1 = \frac{1}{2n^2} - \frac{1}{3n^3} > 0.$$

(Keeping b_1 aside is what makes the final inequality strict; a limit of numbers less than $\frac{1}{n}$ could otherwise equal $\frac{1}{n}$.)

Multiplying by $n > 0$ gives $n\ln(1+\frac{1}{n}) < 1$, that is, $\ln\left((1+\frac{1}{n})^n\right) < 1$. Since the exponential function is increasing,

$$\left(1+\frac{1}{n}\right)^n < \mathrm{e}.$$

Part (ii)

Let $x = \frac{1}{2y}$. Since $y > \frac{1}{2}$, we have $0 < x < 1$, so both x and $-x$ lie in the range where the series is valid:

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots,$$
$$\ln(1-x) = -x - \frac{x^2}{2} - \frac{x^3}{3} - \frac{x^4}{4} - \cdots.$$

Subtracting term by term (legitimate for two convergent series), the even powers cancel and the odd powers double:

$$\ln\left(\frac{1+x}{1-x}\right) = 2\left(x + \frac{x^3}{3} + \frac{x^5}{5} + \cdots\right).$$

Now

$$\frac{1+x}{1-x} = \frac{1+\frac{1}{2y}}{1-\frac{1}{2y}} = \frac{2y+1}{2y-1},$$

so, with $2x = \frac{1}{y}$,

$$\ln\left(\frac{2y+1}{2y-1}\right) = \frac{1}{y} + \frac{2}{3(2y)^3} + \frac{2}{5(2y)^5} + \cdots = \frac{1}{y}\sum_{r=0}^{\infty} \frac{1}{(2r+1)(2y)^{2r}}.$$

Every term after the first is strictly positive (as $y > 0$), so the partial sums from the second onwards are all at least $\frac{1}{y} + \frac{2}{3(2y)^3}$, and hence

$$\ln\left(\frac{2y+1}{2y-1}\right) \geq \frac{1}{y} + \frac{2}{3(2y)^3} > \frac{1}{y},$$

as required.

For the deduction, choose y so that the fraction becomes $1+\frac{1}{n}$. Taking $y = n+\frac{1}{2}$ (which exceeds $\frac{1}{2}$ for every positive integer n),

$$\frac{2y+1}{2y-1} = \frac{2n+2}{2n} = 1+\frac{1}{n},$$

so

$$\ln\left(1+\frac{1}{n}\right) > \frac{1}{n+\frac{1}{2}}.$$

Multiplying by $n+\frac{1}{2} > 0$ gives $\left(n+\frac{1}{2}\right)\ln(1+\frac{1}{n}) > 1$, and exponentiating (an increasing function again),

$$\mathrm{e} < \left(1+\frac{1}{n}\right)^{n+\frac{1}{2}}.$$

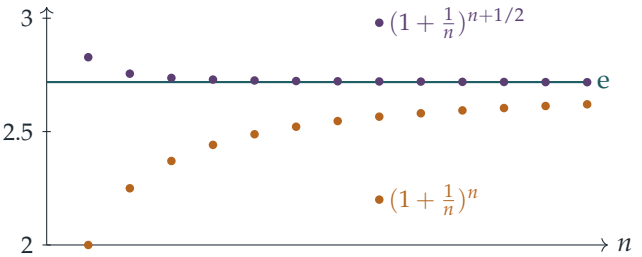
Part (iii)

Dividing the result of part (ii) by $(1+\frac{1}{n})^{1/2} > 0$ and combining it with part (i), for every integer $n \geq 2$

$$\frac{\mathrm{e}}{\sqrt{1+\frac{1}{n}}} < \left(1+\frac{1}{n}\right)^{n} < \mathrm{e}.$$

As $n \to \infty$, $1+\frac{1}{n} \to 1$, so $\sqrt{1+\frac{1}{n}} \to 1$ (the square root is continuous) and the left-hand side tends to e. The right-hand side is the constant e. By the squeeze (sandwich) theorem,

$$\left(1+\frac{1}{n}\right)^{n} \to \mathrm{e}.$$



Checks

For $n = 10$: $(1.1)^{10} \approx 2.5937$ and $(1.1)^{10.5} \approx 2.7203$, which do straddle $\mathrm{e} \approx 2.7183$. For $n = 1000$ the lower bound $\mathrm{e}/\sqrt{1.001} \approx 2.7169237$ is just below $(1.001)^{1000} \approx 2.7169239$, showing how tight the squeeze is. We also checked both strict inequalities for every n up to a few thousand, and checked numerically that the series in part (ii) sums to $\ln\frac{2y+1}{2y-1}$ for random $y > \frac{1}{2}$.

Common slips

- Putting $x = 1$ in part (i): the series is only given for $|x| < 1$, which is why $n > 1$ is needed there (part (iii) only uses large n, so nothing is lost).
- Concluding a strict inequality straight from a limit of strict inequalities. Limits only preserve $\leq$; keep one positive term aside to get $<$.
- Forgetting that exponentiating preserves the inequality only because e^t is increasing; say so.

Beyond STEP

The bounds show more than the limit: since $(1+\frac{1}{n})^{n+1/2}/(1+\frac{1}{n})^n = \sqrt{1+1/n} \approx 1 + \frac{1}{2n}$, the error $\mathrm{e} - (1+\frac{1}{n})^n$ is of size about $\mathrm{e}/(2n)$, so this sequence converges rather slowly. Rearranging convergent series, comparing partial sums and the squeeze theorem are bread and butter in a first analysis course, where you also meet the subtleties (rearranging conditionally convergent series can change the sum!). Abbott's *Understanding Analysis* treats all of this beautifully.

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